

Lunar Surface Infrastructure Program Charter

Educational planning baseline for an integrated lunar demonstration program

Sponsor	Self-directed educational initiative	Status	Planning baseline version 1
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Authorization and purpose

Authorize concept planning for a coordinated lunar surface demonstration that uses robotic systems to prepare and verify a site, supports seven crew members during habitat commissioning, and activates a lunar data center without compromising crew safety. The charter establishes direction, boundaries, governance, measures, and decision gates; it does not represent an approved flight mission or validated engineering design.

Mission objective Demonstrate that robotic deployment, human habitation, shared infrastructure, and computing operations can be integrated and controlled within defined safety, schedule, resource, and performance limits.	Planning assumptions Seven crew members; 45-day nominal surface mission; consumables sized for at least 60 days; illuminated south-polar ridge as the reference site; robotic assets arrive before the crew; essential infrastructure is remotely verified before crew launch.
Program scope 1 Robotic exploration and infrastructure deployment 2 Seven-person habitat commissioning mission 3 Lunar data-center commissioning and operations Shared systems include power, thermal control, communications, logistics, cybersecurity, safety, and mission integration.	Operating priorities 1 Protect crew life and safe departure capability 2 Preserve habitat and critical communications 3 Complete robotic site and infrastructure objectives 4 Deliver critical computing 5 Execute discretionary science and computing
Out of scope Flight-certified design, launch-vehicle selection, procurement commitments, detailed cost certification, medical certification, resource ownership claims, and representation of simulated authority as real-world program experience.	Program leadership Program Director owns the integrated baseline, priorities, resources, risk posture, readiness decisions, and executive communication. Project leads own delivery within approved scope. The Chief Engineer owns technical integration; Safety and Mission Assurance retains independent stop-work authority.

Success measures

Mission readiness	Crew and habitat	Deployment	Data center
All gate-entry criteria met; no unaccepted critical risk; integrated schedule and interfaces baselined.	No loss of life-support capability; survival reserves remain above approved thresholds; safe departure preserved.	Robotic survey, site preparation, infrastructure placement, and remote verification completed to plan.	Critical computing is commissioned within approved power and thermal limits; noncritical loads shed before survival loads.

Decision gates

G1 Mission baseline	G2 Site readiness	G3 Crew launch	G4 Data-center start	G5 Mission closeout
Approve objectives, requirements, organization, funding assumptions, interfaces, and integrated schedule.	Accept survey results, landing hazards, site preparation, and infrastructure-placement evidence.	Verify habitat pressure, life support, power, communications, supplies, shelter, and departure capability.	Confirm stable habitat operations and adequate power, thermal, communications, and emergency margins.	Accept mission data, residual risk, asset configuration, lessons learned, and follow-on recommendations.

Initial risks and controls

Risk	Potential program effect	Initial control approach
Shared power or thermal shortfall	Competition between habitat and computing reduces survival reserve or mission output.	Protected survival bus, load-shedding hierarchy, independent emergency power, margin reviews.
Robotic deployment failure	Site remains unverified or infrastructure is unavailable before crew launch.	Redundant assets, remote inspection, repair capability, and no-go launch gate.
Natural hazard or delayed departure	Radiation, dust, impact, eclipse, or vehicle issue extends exposure and consumption.	Storm shelter, reserves, fault isolation, contingency schedule, and departure-readiness monitoring.

Immediate next decision Approve development of the Level 1 work breakdown structure, integrated milestone schedule, responsibility matrix, requirements baseline, and initial risk and opportunity register.

Reference basis NASA lunar south-pole candidate-region criteria and NASA human-spaceflight hazard framework. All mission figures remain educational planning assumptions pending analysis.